

Andrew Estrada

andrewgestrada@gmail.com | andrewgestrada.com | [linkedin.com/in/andrew-estrada2024/](https://www.linkedin.com/in/andrew-estrada2024/)

Summary: Mechanical/Manufacturing Engineer with strong shop-floor & CNC background: Haas mills/lathes, Mastercam. CAD/CAE (SolidWorks, ANSYS); GD&T, DFM, Lean/6S, and WI/SOPs. Led NAU FSAE front pushrod suspension & test (Lotus Shark, DAQ). NASA Student Launch avionics/GPS and recovery tuning.

EDUCATION

Northern Arizona University

B.S. Mechanical Engineering

Aug 2020 - May 2024

Flagstaff, AZ

- NAU Founders Gold Scholarship
- Member of Formula SAE, SAE Baja, Space Jacks and ASME

RELEVANT WORK EXPERIENCE

Strategic Enrollment & Marketing- NAU | *Digital Marketing Assistant* | *Remote*

Nov 2023 - Present

- Served on the development team for new website, enhancing UX, performance, and backend workflows
- Conducted SEO analysis, implementing strategies to enhance search visibility and organic traffic
- Developed and utilized custom tools in Python, leveraged REST APIs for data extraction and performance analytics
- Researched and fine-tuned LLMs applying NLP techniques to create Q&A datasets and develop AI productivity tools

Parker Meggitt | *Manufacturing Engineering Intern* | *Simi Valley, CA*

July 2023 - Aug 2023

- Designed and implemented innovative fixtures to enhance process safety and ergonomics, increasing efficiency, productivity, and adherence to Lean manufacturing
- Collaborated with design engineers to conduct root cause analysis and implement effective corrective actions
- Provided invaluable assistance to production personnel by updating and clarifying work instructions (WI/SOPs), optimizing production processes, and ensuring smooth operations

Accu-Glass Products, Inc. | *Engineering Intern* | *Valencia, CA*

May 2022 - Aug 2022

- Operated Haas CNC 3-axis Mills and 2/4-axis Lathes to complete components for capital projects
- Created CAD models, technical drawings for products, supported work holding design and manufacturing processes
- Initiated precision quality control and effectively took precise measurements to ensure effective tolerance for high-accuracy air to high and ultra-high vacuum parts

Formula SAE | *Northern Arizona University*

Apr 2023 - May 2024

- Front Pushrod Suspension Lead: responsible for design, validated suspension geometry within Lotus Shark, utilized ANSYS to simulate driving conditions, collaborated with chassis team for successful integration
- Manufactured Front Pushrod Assembly utilizing CNC Mill and Lathe, programmed with Mastercam
- Team Test Engineer: responsible for test planning, data acquisition and analysis (DAQ), evaluating design choices and vehicle dynamics against lap simulations to suggest car setup changes
- Co-founder: managed sponsor relations, new member outreach, website development, and social media presence

NASA Student Launch Competition | *Northern Arizona University*

Sept 2022 - Apr 2023

- Designed and tested GPS subsystem, assisted with avionics integration, programming, and troubleshooting
- Conducted rocket performance simulations in OpenRocket to validate flight dynamics and optimize stability
- Analyzed and optimized stage ejection methodology for a reliable recovery system
- Served as safety officer, ensuring compliance with NASA competition regulations and enforcing safety protocols

Eagle Scout Project | *Haskell Canyon Open Space*

Jul 2018 - Dec 2019

- Initiated and led a year-long community service project in partnership with the City of Santa Clarita, California, to renovate and enhance Haskell Canyon Open Space. Designed a kiosk, benches, and trail signage using CAD; secured funding through targeted business presentations; and coordinated 50+ volunteers for successful construction and installation

SKILLS & INTERESTS

- **Technical Skills:** SolidWorks; ANSYS; MATLAB; Mastercam ; Python; Fusion 360; Inventor; GD&T; FMEA
- **Manufacturing Skills:** Haas 3-Axis CNC Mill and 2 & 4-Axis Lathe Experience; FDM 3D Printing; LSS
- **Interests:** Mountain Biking; High Power Rocketry; Sound Systems and Acoustics; Cars; Backpacking; Travel